

1. Administrative & Study Identification

Full Study Title: Effectiveness of BNT162b2 Formulations Using State Vaccine Registry and Insurance Claims Data **Short Title/ Acronym:** RAVEN
Protocol Number: C4591065 **NCT Number:** NCT06199934
Sponsor Name: Pfizer
Investigational Product: BNT162b2 (Pfizer-BioNTech COVID-19 Vaccine / XBB.1.5-adapted formulation)
Phase of Development: N/A (Observational / Retrospective Cohort Study)
Medical Monitor: Pfizer Study Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate how well different versions of the Pfizer-BioNTech COVID-19 vaccine (BNT162b2) work at preventing death, severe COVID-19 requiring hospitalization, and overall use of healthcare resources (e.g., emergency department or urgent care visits). **Secondary Objectives:** To estimate adjusted vaccine effectiveness in specific subpopulations (such as children aged 5-17 years) using time-varying exposure definitions.

2.2 Background & Rationale To assess the real-world effectiveness of updated BNT162b2 formulations, including the XBB.1.5-adapted vaccine. As the virus continues to evolve, it is necessary to leverage existing state vaccine registries and insurance claims data (such as California and Louisiana datasets via HealthVerity) to continuously evaluate the vaccine's protective effect against severe COVID-19 outcomes and healthcare utilization.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective Cohort Study) **Control Type:** Unvaccinated or prior formulation (comparison of vaccinated vs. unvaccinated person-time) **Blinding:** N/A (Open-label / Database Study) **Randomization:** N/A (Non-probability sample)

3.2 Planned Enrollment & Duration Target \$N\$: 19,853,610 participants **Number of Sites:** N/A (Real-world data from US states: California and Louisiana) **Estimated Study Start:** 25-Sep-2023 (Observation window start) **Estimated Primary Completion:** 31-Mar-2024 (Observation window end)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. People with at least one year of pharmacy and medical enrollment in HealthVerity prior to vaccine availability. 2. State of California or Louisiana resident for at least one year.

4.2 Exclusion Criteria 1. People currently pregnant. 2. Individuals with discrepancies in sex and/or year of birth between HealthVerity claims and California/Louisiana immunization registry datasets. 3. A diagnosis of COVID-19 in any setting ≤ 90 days prior to the start of the study or receipt of a COVID-19 vaccine ≤ 90 days prior to the start of the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of BNT162b2 (e.g., XBB.1.5-adapted formulation) or previous doses as recorded in the registry. **Route of Administration:** Intramuscular (IM) (Standard for BNT162b2). **Duration of Treatment:** Single administration, with follow-up evaluating vaccinated time ≥ 14 days post-vaccination.

5.2 Concomitant & Prohibited Therapy Permitted: N/A (Retrospective real-world study). **Prohibited:** N/A (Retrospective real-world study).

6. Study Timeline & Visit Schedule

(Note: As this is a retrospective database study, physical visits are not applicable. Schedule represents data analysis points)

Visit ID	Window (Days)	Key Activities/Procedures
Screening	N/A	Database search for eligibility based on at least 1 year of HealthVerity enrollment and state residency
Baseline	Day 0 (Index)	Tokenized linkage of claims to state registries (California and Louisiana)
Interim	N/A	Longitudinal tracking of COVID-19 incidence, ED/UC visits, and hospitalizations
End of Study	Follow-up cut-off	Final data extraction and time-dependent Cox model analysis

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of participants diagnosed with COVID-19, including severe COVID-19 requiring hospitalization, and prevention of death or overall healthcare resource utilization. **Measurement Method:** Incidence rates and adjusted hazard ratios calculated via time-dependent Cox models using HealthVerity claims tokenized to state vaccine registries.

7.2 Secondary & Exploratory Endpoints Adjusted vaccine effectiveness (calculated as $[1 - \text{Hazard Ratio}] \times 100\%$) against COVID-19-associated ED/UC encounters. Vaccine uptake and effectiveness stratified by age and prior vaccination status.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Assessed via medical diagnostic and ICD codes in HealthVerity claims data. **Serious Adverse Events (SAEs):** Severe COVID-19 outcomes, hospitalization, and death extracted from linked healthcare records. **Data Monitoring Committee (DMC):** N/A (Retrospective observational study).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Retrospective cohort of eligible residents (overall target ~19.8M; e.g., pediatric subgroup of ~2.45M). **Sample Size Justification:** Based on available real-world non-probability samples in HealthVerity linked with California/Louisiana registries. **Handling of Missing Data:** Exclusion of subjects with discrepancies in sex and/or year of birth between claims and immunization registries.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with real-world data regulations and guidelines for retrospective observational research, ensuring patient privacy and HIPAA compliance via tokenization. **Monitoring Plan:** N/A (Retrospective data extraction and tokenization). **Audit Trail:** Tokenized linkage algorithm logs between HealthVerity claims and state registries.

11. Study Identifiers & Governance

Primary ID: C4591065 **Registry Number:** NCT06199934 **Sponsor/Lead Organization:** Pfizer **Study Title:** RAVEN **Official Regulatory Title:** Effectiveness of BNT162b2 Formulations Using State Vaccine Registry and Insurance Claims Data

12. Clinical Framework (COVID-19 Vaccine Specific)

Primary Condition: COVID-19 Infection and Severe Outcomes **Diagnosis Threshold:** Documented diagnosis of COVID-19 in claims data / healthcare settings **Medical Methodology:** BNT162b2 Vaccination (including XBB.1.5-adapted formulation) **Optimization Target:** Prevention of death, hospitalization, and ED/UC encounters

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Pfizer Study Lead]	_____	[DD-MMM-YYYY]					
	Clinical Operations Lead	[Name]	_____					
	Lead Statistician	[Name]	_____					